

DAY 3 – PYTHON SPRINT

Core Topic: Conditional Statements

Resource: <https://www.youtube.com/watch?v=UrsmFxEIp5k> from 3hr 09 minutes till 3hrs 58 minutes

[Expected Time to Finish – 2 hours]

Core Problem: Daily Expense Tracker

Everyone works on the same problem. Difficulty depends on how deep you take it. Choose your level and complete it properly.

BEGINNER LEVEL (Starting from Scratch)

Write a Python program that does the following:

1. Take **user name** as input.
2. Take names and prices of **3 products** from the user.
3. Store product–price pairs in a **dictionary**.
4. Calculate the **total cart value**.
5. Use if / elif / else to apply discount:
 - $\geq 5000 \rightarrow 20\%$ discount
 - $\geq 2000 \rightarrow 10\%$ discount
 - else $\rightarrow$ No discount
6. Calculate the **final payable amount**.
7. Print a **clean cart summary** (user, products & prices, total, discount, final amount).

Expected Time to Finish – 20 min

Sample Input:

```
Enter username: Arshad Ali
Enter product 1 name: Biscuits
Enter product 1 price: 50
Enter product 2 name: Laptop
Enter product 2 price: 80000
Enter product 3 name: Phone
Enter product 3 price: 8000
```

Sample Output:

```
----- CART SUMMARY -----
User: Arshad Ali
Products in Cart:
Biscuits: 50
Laptop: 80000
Phone: 8000
Total Amount: 88050
Discount: 20%
Final Amount: 70440.0
```

INTERMEDIATE LEVEL

Instructions – Smart Shopping Cart Analyzer

- Create a **predefined dictionary** of products where each product has:
 - name
 - price
 - category

❓ Take **user name** as input.

❓ Display the **product menu** with numbers (1–4).

❓ Ask the user to **select 3 products using numbers**.

❓ Using the selected numbers:

- Store product names in a **list**
 - Store product–price pairs in a **dictionary**
 - Store unique categories in a **set**
- Ask the user for **payment method** (UPI, Card, COD).
- Calculate **total cart value**.
- Apply discount using if / elif / else:
 - $\geq 5000 \rightarrow 20\%$
 - $\geq 2000 \rightarrow 10\%$
 - else $\rightarrow 0\%$
- Apply payment rules:
 - COD + Electronics $\rightarrow$ Order not allowed
 - Card + total $> 30000 \rightarrow$ Extra 5% discount
- Calculate **final amount** and print a **purchase summary**.

Sample Input:

```
Enter user name: Arshad-Ali
Available Products:
1. Laptop - ₹45000 (Electronics)
2. Mobile - ₹20000 (Electronics)
3. Rice - ₹1200 (Grocery)
4. Shirt - ₹1800 (Clothing)
Select product 1 (1-4): 1
Select product 2 (1-4): 2
Select product 3 (1-4): 2
Enter payment method (UPI/Card/COD): COD
```

Sample Output:

```
=====
                PURCHASE SUMMARY
=====
User           : Arshad-Ali
Products       : ['Laptop', 'Mobile', 'Mobile']
Categories     : {'Electronics'}
Payment        : COD

Cart Total     : 85000
Discount       : 20%
Final Price    : 68000.0
Status         : Order Not Allowed (COD + Electronics)
=====
```

ADVANCED LEVEL MINI PROJECT



SMART CLI SHOPPING SYSTEM

Python Mini-Project (Intermediate Level)



Objective

Build a **menu-driven, CLI-based shopping system** in Python that simulates a real-world checkout flow with:

- Admin-controlled product catalog
 - Customer shopping cart
 - Category-based business rules
 - Payment and discount logic
 - Final purchase summary
-



Part 1: Product Catalog (Admin Controlled)

Requirements

- Store products in a dictionary.
- Each product must have:
 - Product ID (integer key)
 - Name
 - Price
 - Category

Categories (Fixed)

- Electronics
 - Grocery
 - Clothing
-



Part 2: Admin Panel

Features

Admin should be able to:

1. View all available products
2. Add a new product
3. Prevent duplicate product IDs
4. Exit back to main menu

Rules

- Product IDs must be unique
 - Price must be numeric
 - Category must be one of the allowed categories
-



Part 3: Customer Panel

Flow

1. Ask customer name
 2. Display available products
 3. Allow user to:
 - Add items using Product ID
 - Select quantity
 4. Allow checkout using keyword `done`
 5. Maintain:
 - Cart
 - Total price
 - Product categories purchased
-



Part 4: Payment System

Allowed Payment Methods

- UPI
 - Card
 - COD
-



Part 5: Discount Rules

Cart Total (₹) Discount

≥ 5000	20%
≥ 2000	10%
< 2000	0%

Bonus Rule

- If payment method is **Card** and total $> ₹30,000 \rightarrow +5\%$ extra discount

Part 6: Business Restrictions

Mandatory Rule

- **COD is NOT allowed for Electronics**
 - If violated → Order status = Order Not Allowed
-



Part 7: Purchase Summary

At checkout, display:

- Customer name
- Itemized cart
- Categories purchased
- Payment method
- Cart total
- Discount percentage
- Final payable amount
- Order status

Formatting and clarity matter.



Validation Requirements

- Handle invalid menu choices
- Handle invalid product IDs
- Prevent empty cart checkout
- Gracefully handle wrong inputs

Final Note

This is **not a beginner task**.

Treat it like a **real system design problem**, not a syntax exercise.

Sample Run:

```

=====
SMART SHOPPING SYSTEM
=====

1. Admin Panel
2. Customer Panel
3. Exit
Choose an option: 1

--- ADMIN PANEL ---

1. View Products
2. Add Product
3. Exit Admin Panel
Choose an option: 2
Enter new product ID: 5
Enter product name: Headphones
Enter product price: 2500
Enter category (Electronics/Grocery/Clothing): Electronics
Product added successfully!

Choose an option: 3

=====
SMART SHOPPING SYSTEM
=====

1. Admin Panel
2. Customer Panel
3. Exit
Choose an option: 2

--- CUSTOMER PANEL ---
Enter your name: Arshad

Available Products:
1 - Laptop | ₹ 45000 | Electronics
2 - Mobile | ₹ 20000 | Electronics
3 - Rice | ₹ 1200 | Grocery
4 - Shirt | ₹ 1800 | Clothing
5 - Headphones | ₹ 2500 | Electronics

```

```

Enter product ID to add (or 'done' to checkout): 1
Enter quantity: 1

Enter product ID to add (or 'done' to checkout): 3
Enter quantity: 2

Enter product ID to add (or 'done' to checkout): done

Enter payment method (UPI/Card/COD): Card

=====
PURCHASE SUMMARY
=====
Customer : Arshad

Items:
Laptop x 1 → ₹ 45000
Rice x 2 → ₹ 2400

Categories : {'Electronics', 'Grocery'}
Payment : Card

Cart Total : 47400
Discount : 25%
Final Pay : 35550.0
Status : Order Confirmed

=====

SMART SHOPPING SYSTEM
=====

1. Admin Panel
2. Customer Panel
3. Exit
Choose an option: 3
Thank you for using the system!

```

Submission & Deadlines

Deadline: Tomorrow by 9:15 PM.

Format: Submit via WhatsApp using a link to GitHub !!, or by pasting the code (not recommended)